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# Malicious Code Analysis

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Part One

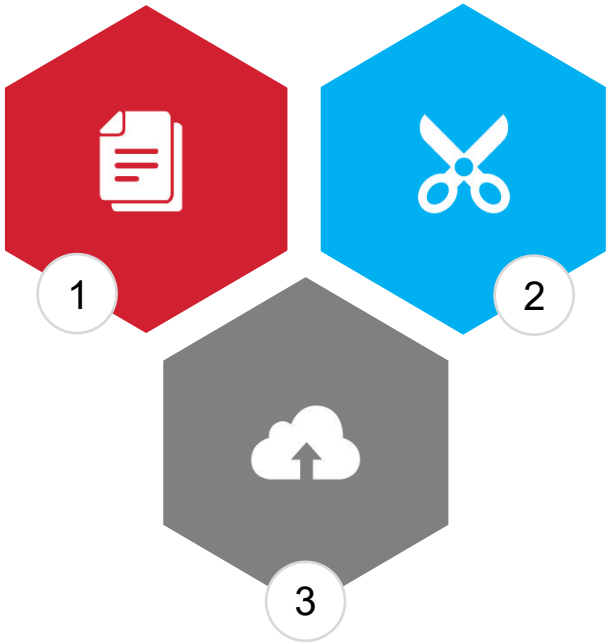
01

# Dynamic Malware Analysis

2025-11-14

# What is Dynamic Analysis?

- Allow the malware to run.




- Complement static analysis.

- Help us to understand the program behavior.

# Dynamic Analysis

**Basic dynamic analysis**

- Executing target binary and observing its behavior.

**Advanced dynamic analysis**

- Executing target binary and using a debugger to analyze internal states.

# Executing the Malware

- In most cases, you can just double-click the “exe” file.
  - You may want to run it from the command-line as well.
- What if the extension is not “exe”?
  - You can change it. Verify if it is a PE file using a PE parser.
- What if it is a DLL?
  - You can run a DLL using the rundll32 program.
  - Format: C:\> rundll32.exe <name>.dll

# Dynamic Analysis Systems

- Dynamic analysis is run in a safe environment on dedicated physical or virtual machines (in order not to expose the users' system to unnecessary risks)
  - ***Physical machines*** are set up on isolated networks, disconnected from the Internet or any other network, to prevent malware from spreading
  - ***Virtual machines*** emulate the functionality of a physical computer, where the OS running on the virtual machine is isolated from the host OS
    - *One limitation is that some malware can detect when they are running in a virtual machine, and they will execute differently than when in a physical machine*
  - A related term is ***sandbox***, referring to a physical or virtual environment for running malware, which isolates executables from other system resources and applications.
    - *Although they share characteristics with physical and virtual machines, sandboxes can be more limited (e.g., they can run in the browser), while physical and virtual machines always act as a complete system*
    - *For example, online sandboxes are websites where one can submit a sample file and receive a report about its behavior*

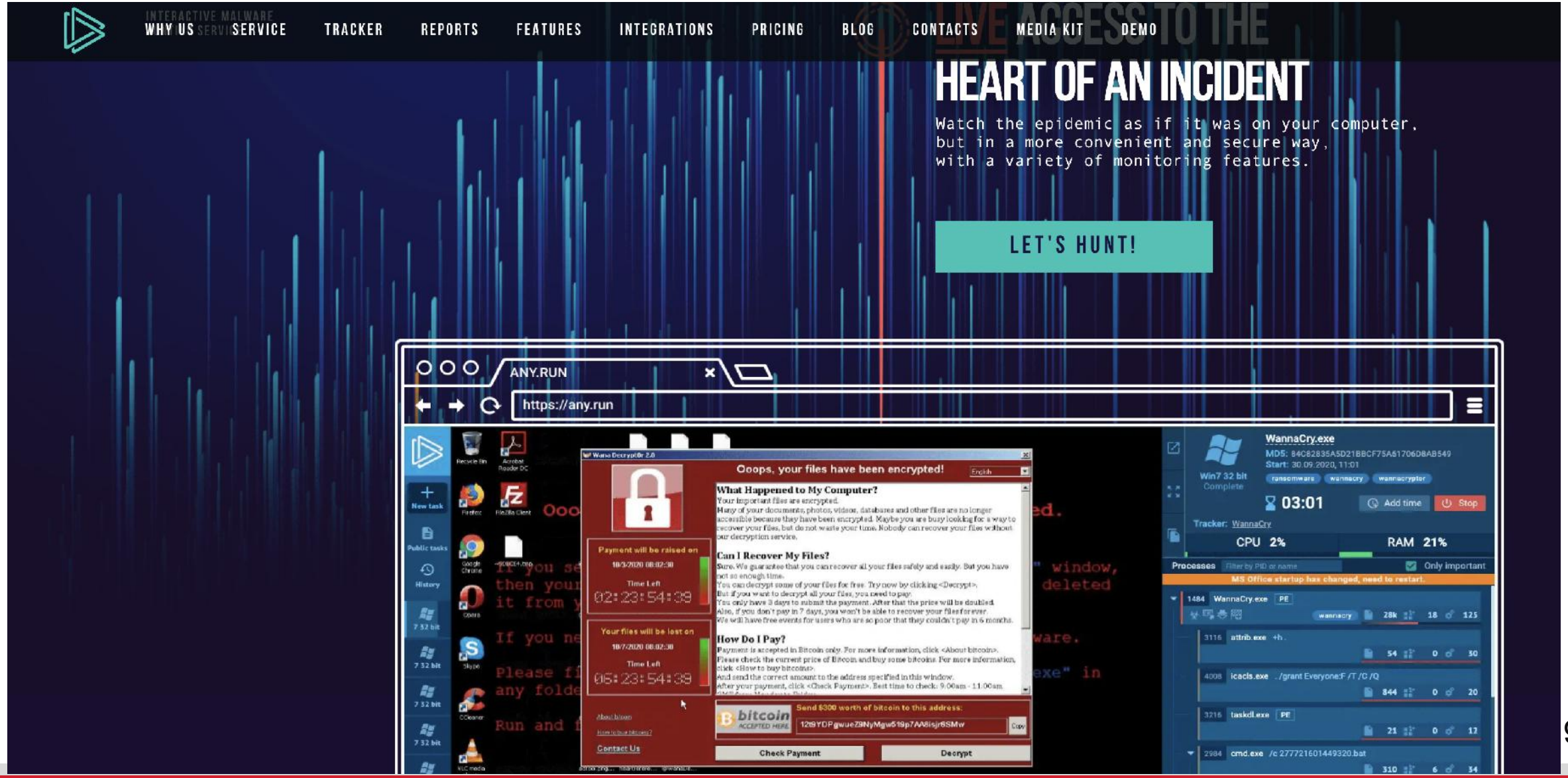
# Sandboxes

- All-in-one software solutions to analyze the execution of a program.
- Provides security mechanisms for running untrusted programs in a safe environment.
- Let you moniter behavior/changes to the system.
- The “real” system remains isolated-so, it does not get infected.

# Sandboxes-drawbacks

- May run the EXE w/o command line arguments.
- Execution may wait for response from C2.
- Malware may find out that it is running in a sandbox .
  - an anti-analysis technique
  - in that case, the malware may change its behavior
- Often the environment is not properly setup.

# Sandbox-Example:https://any.run/



# Dynamic Features for Malware Classification

- ***Dynamic features*** are extracted from the execution of malware at runtime
  - **Memory and registers usage -** values stored in the memory and different registers during the execution can distinguish benign from malicious programs
  - **Instruction traces -** sequence of processor instructions called during the execution of a program

# Dynamic Features for Malware Classification

- **Network traffic -** monitoring the traffic entering and exiting the network can provide helpful information to detect malicious behavior
- **API call traces -** traces for accessing file systems, devices, processes, threads and error handling, and also to access functions such as the Windows registry, manage user accounts, etc.

# Moniter System Activity

## Process Moniter

- Allow monitering of registry, file system, network, process, and thread activities.
- Moniter all system calls
- Captures a lot of data (>50,000 events in a minute)
- Use RAM to capture events
  - can easily crash a VM - so, run for a limited amount of time
- Not a reliable tool for network activities
  - so other tools needs to be used

# Running Malware-Process Moniter

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

| Time ... | Process Name | PID | Operation | Path | Result | |
|---|---|---|---|---|---|---|
| 1:20:1... | svchost.exe | 2152 | ReadFile | C:\Windows\System32\StateRepository... | SUCCESS | C |
| 1:20:1... | MsMpEng.exe | 5060 | ReadFile | C:\ProgramData\Microsoft\Windows De... | SUCCESS | C |
| 1:20:1... | svchost.exe | 2152 | ReadFile | C:\Windows\System32\StateRepository... | SUCCESS | C |
| 1:20:1... | MsMpEng.exe | 5060 | ReadFile | C:\ProgramData\Microsoft\Windows De... | SUCCESS | C |
| 1:20:1... | svchost.exe | 2152 | ReadFile | C:\Windows\System32\StateRepository... | SUCCESS | C |
| 1:20:1... | svchost.exe | 2152 | ReadFile | C:\Windows\System32\Windows.State... | SUCCESS | C |
| 1:20:1... | MsMpEng.exe | 5060 | ReadFile | C:\ProgramData\Microsoft\Windows De... | SUCCESS | C |
| 1:20:1... | lsass.exe | 1232 | ReadFile | C:\Windows\System32\lsasrv.dll | SUCCESS | C |
| 1:20:1... | svchost.exe | 2152 | ReadFile | C:\Windows\System32\Windows.State... | SUCCESS | C |
| 1:20:1... | lsass.exe | 1232 | ReadFile | C:\Windows\System32\lsasrv.dll | SUCCESS | C |
| 1:20:1... | MsMpEng.exe | 5060 | ReadFile | C:\ProgramData\Microsoft\Windows De... | SUCCESS | C |
| 1:20:1... | lsass.exe | 1232 | ReadFile | C:\Windows\System32\lsasrv.dll | SUCCESS | C |
| 1:20:1... | MsMpEng.exe | 5060 | ReadFile | C:\ProgramData\Microsoft\Windows De... | SUCCESS | C |
| 1:20:1... | svchost.exe | 2152 | UnlockFileSingle | C:\ProgramData\Microsoft\Windows\A... | SUCCESS | C |
| 1:20:1... | svchost.exe | 2152 | ReadFile | C:\Windows\System32\StateRepository... | SUCCESS | C |
| 1:20:1... | lsass.exe | 1232 | QueryNameInfo... | C:\Users\defaultuser0.DESKTOP-931H... | SUCCESS | N |
| 1:20:1... | lsass.exe | 1232 | QueryNameInfo... | C:\Users\defaultuser0.DESKTOP-931H... | SUCCESS | N |
| 1:20:1... | MsMpEng.exe | 5060 | ReadFile | C:\Users\defaultuser0.DESKTOP-931H... | SUCCESS | C |
| 1:20:1... | explorer.exe | 6452 | ReadFile | C:\Windows\System32\SHCore.dll | SUCCESS | C |
| 1:20:1... | svchost.exe | 2152 | ReadFile | C:\Windows\System32\Windows.State... | SUCCESS | C |
| 1:20:1... | MsMpEng.exe | 5060 | ReadFile | C:\Users\defaultuser0.DESKTOP-931H... | SUCCESS | C |
| 1:20:1... | explorer.exe | 6452 | ReadFile | C:\Windows\System32\Taskbar.dll | SUCCESS | C |
| 1:20:1... | MsMpEng.exe | 5060 | ReadFile | C:\ProgramData\Microsoft\Windows De... | SUCCESS | C |
| 1:20:1... | MsMpEng.exe | 5060 | ReadFile | C:\Users\defaultuser0.DESKTOP-931H... | SUCCESS | C |
| 1:20:1... | svchost.exe | 2152 | LockFile | C:\ProgramData\Microsoft\Windows\A... | SUCCESS | E |
| 1:20:1... | svchost.exe | 2152 | QueryStandardI... | C:\ProgramData\Microsoft\Windows\A... | SUCCESS | A |
| 1:20:1... | svchost.exe | 2152 | UnlockFileSingle | C:\ProgramData\Microsoft\Windows\A... | SUCCESS | C |
| 1:20:1... | explorer.exe | 6452 | ReadFile | C:\Windows\System32\Taskbar.dll | SUCCESS | C |
| 1:20:1... | MsMpEng.exe | 5060 | ReadFile | C:\Users\defaultuser0.DESKTOP-931H... | SUCCESS | C |
| 1:20:1... | MsMpEng.exe | 5060 | ReadFile | C:\ProgramData\Microsoft\Windows De... | SUCCESS | C |
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| 1:20:1... | svchost.exe | 2152 | UnlockFileSingle | C:\ProgramData\Microsoft\Windows\A... | SUCCESS | C |
| 1:20:1... | MsMpEng.exe | 5060 | CloseFile | C:\Users\defaultuser0.DESKTOP-931H... | SUCCESS | |
| 1:20:1... | MsMpEng.exe | 5060 | ReadFile | C:\ProgramData\Microsoft\Windows De... | SUCCESS | C |
| 1:20:1... | MsMpEng.exe | 5060 | CloseFile | C:\Users\defaultuser0.DESKTOP-931H... | SUCCESS | |
| 1:20:1... | explorer.exe | 6452 | RegQueryKey | HKCU\Software\Classes | SUCCESS | C |
| 1:20:1... | explorer.exe | 6452 | RegQueryKey | HKCU\Software\Classes | SUCCESS | C |
| 1:20:1... | explorer.exe | 6452 | RegQueryKey | HKCU\Software\Classes | SUCCESS | C |
| 1:20:1... | explorer.exe | 6452 | RegOpenKey | HKCU\Software\Classes\Applications\... | NAME NOT FOUND | D |
| 1:20:1... | explorer.exe | 6452 | RegOpenKey | HKCR\Applications\Procmon64.exe | NAME NOT FOUND | D |
| 1:20:1... | explorer.exe | 6452 | ReadFile | C:\Windows\System32\windows.storag... | SUCCESS | C |
| 1:20:1... | explorer.exe | 6452 | RegQueryKey | HKCU\Software\Classes | SUCCESS | Query: Name |
| 1:20:1... | explorer.exe | 6452 | RegQueryKey | HKCU\Software\Classes | SUCCESS | Query: HandleTag... |
| 1:20:1... | explorer.exe | 6452 | RegQueryKey | HKCU\Software\Classes | SUCCESS | Query: Name |
| 1:20:1... | explorer.exe | 6452 | RegOpenKey | HKCU\Software\Classes\Applications\... | NAME NOT FOUND | Desired Access: R... |
| 1:20:1... | explorer.exe | 6452 | RegOpenKey | HKCR\Applications\Procmon64.exe | NAME NOT FOUND | Desired Access: R... |
| 1:20:1... | svchost.exe | 2152 | LockFile | C:\ProgramData\Microsoft\Windows\A... | SUCCESS | Exclusive: False, O... |

Event Properties

Event | Process | Stack

Image

Host Process for Windows Services
Microsoft Corporation

Name: svchost.exe
Version: 10.0.22621.1 (WinBuild.160101.0800)

Path:
C:\WINDOWS\system32\svchost.exe

Command Line:
C:\WINDOWS\system32\svchost.exe -k appmodel -p -s StateRepository

| | | | |
|---|---|---|---|
| PID: | 2152 | Architecture: | 64-bit |
| Parent PID: | 1192 | Virtualized: | False |
| Session ID: | 0 | Integrity: | System |
| User: | NT AUTHORITY\SYSTEM | | |
| Auth ID: | 00000000:000003e7 | | |
| Started: | 3/24/2023 4:44:49 PM | Ended: | (Running) |

Modules:

| Module | Address | Size | Path | Company | Version | Ti |
|---|---|---|---|---|---|---|
| svchost.exe | 0x7ff62eee0000 | 0x11000 | C:\WINDOWS\system32\svchost.... | Microsoft Cor... | 10.0.22621.1 (... | 5/ |
| windows.stater... | 0x7fff1cde0000 | 0x3d000 | C:\WINDOWS\SYSTEM32\window... | Microsoft Cor... | 10.0.22621.457 ... | 1 |
| Windows.State... | 0x7fff25750000 | 0xeb000 | C:\Windows\System32\Windows.... | Microsoft Cor... | 10.0.22621.457 ... | 6/ |
| windows.stater... | 0x7fff27190000 | 0x1a000 | C:\WINDOWS\SYSTEM32\window... | Microsoft Cor... | 10.0.22621.457 ... | 4/ |
| StateRepository... | 0x7fff27250000 | 0xb3000 | c:\windows\system32\StateRepos... | Microsoft Cor... | 10.0.22621.457 ... | 1 |
| windows.stater... | 0x7fff27380000 | 0x682000 | c:\windows\system32\windows.st... | Microsoft Cor... | 10.0.22621.1 (... | 1 |

Next Highlighted | Copy All | Close

# Running Malware-Process Moniter

- To narrow the result, use filtering
- You may want to filter on:

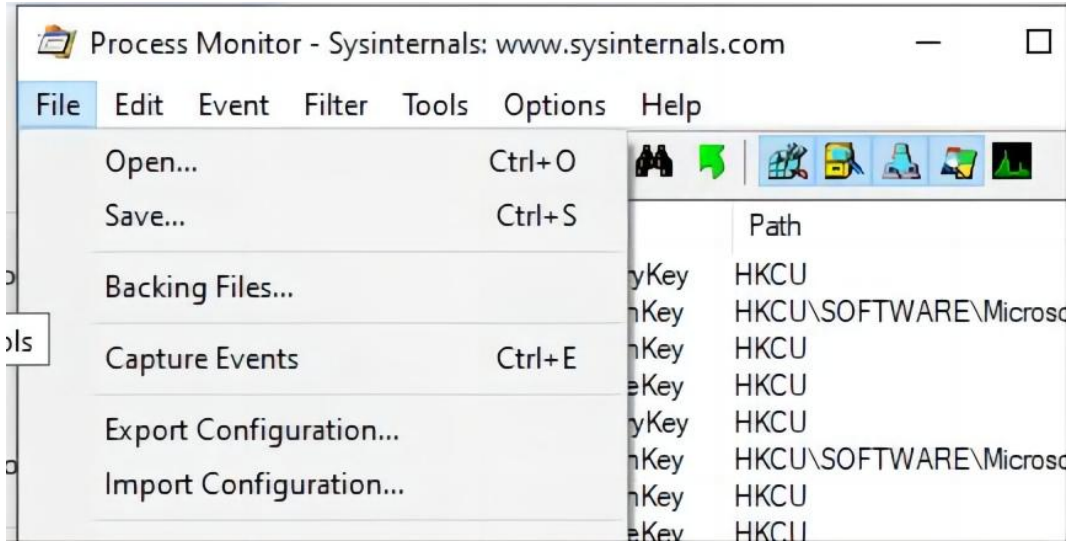
  Executables running on the system
  System call (such as RegSetValue, Create File, WriteFile, etc.)

**Note:** Filtering does not prevent from consuming too much memory though.

# Example1-Track File and Registry Changes

- Let's say, you need to track access to the registry key HKEY_CURRENT_USER\Software\test and file c:\ps\procmon_example.txt.
- When Process Monitor starts, it begins capturing all events according to the default filters.

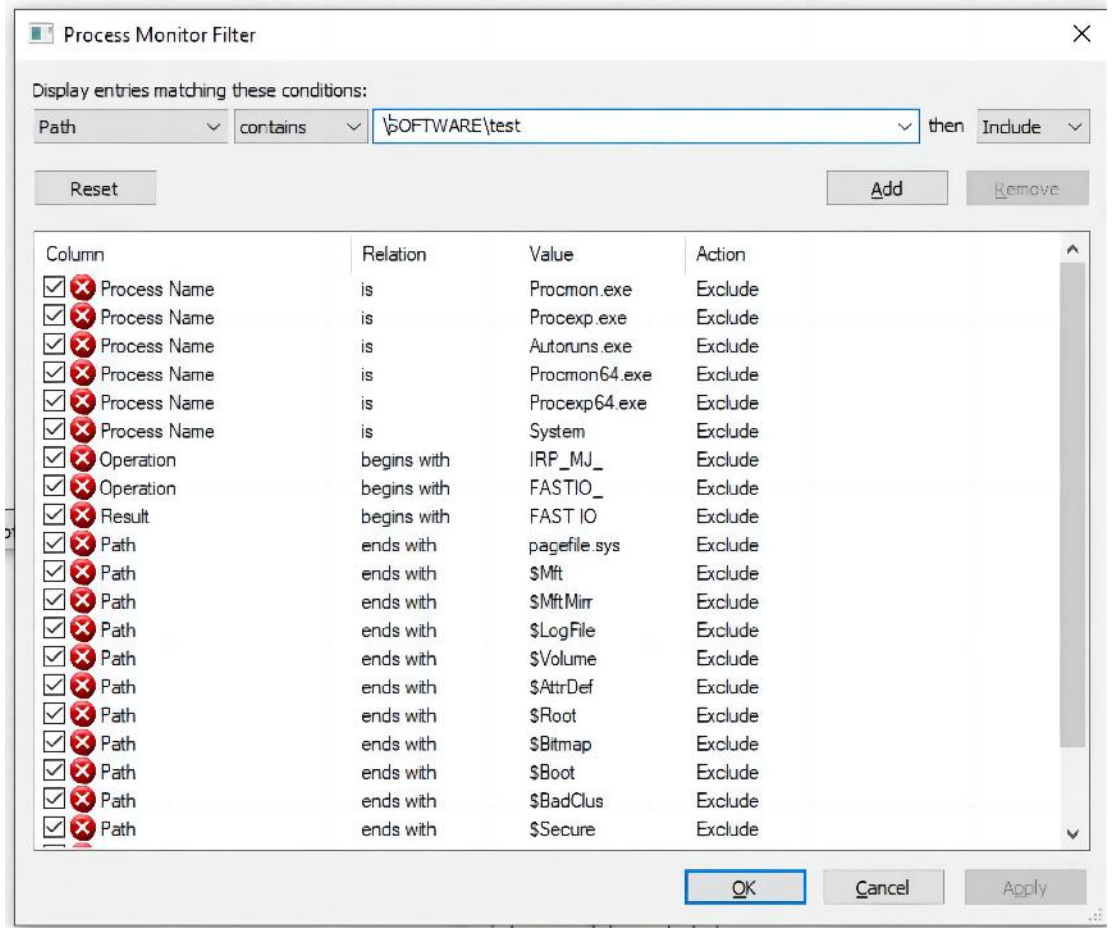
- Step 1. Stop capturing events by unchecking the option File > Capture Events (Ctrl+E).
- Step 2. Clear the current ProcMon log (Edit > Clear Display).



# Example1-Track File and Registry Changes

Let's say, you need to track access to the registry key HKEY_CURRENT_USER\Software\test and file c:\ps\procmon_example.txt.

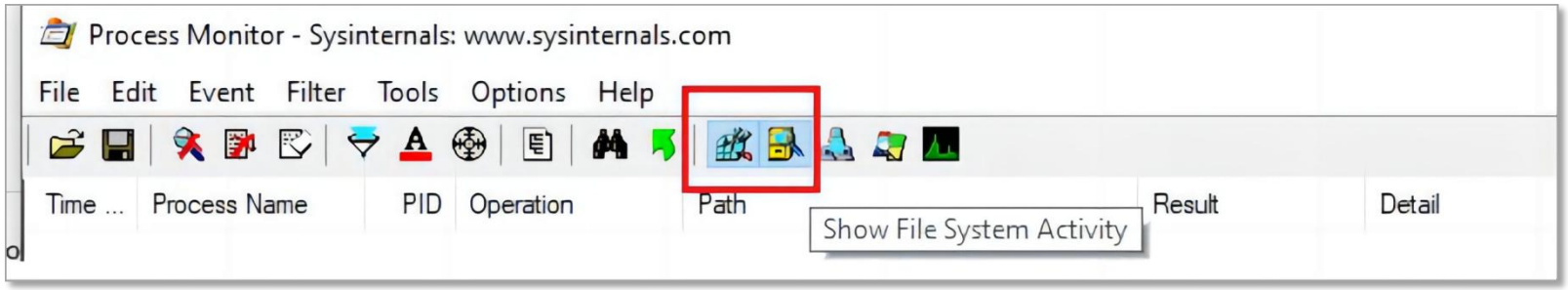
- Step 3. Now you need to configure the Process Monitor filters (Filter > Filter).
- Step 4. Create a filter for monitoring access to the registry key: Path > contains > \SOFTWARE\test > Include.
- Step 5. Click Add to add a new filter to the list



# Example1-Track File and Registry Changes

Let's say, you need to track access to the registry key HKEY_CURRENT_USER\Software\test and file c:\ps\procmon_example.txt.

- Step 6. Now add a file access event filter: Path > is > c:\ps\procmon_example.txt > Include.
- Step 7. Make sure the following options are enabled in the toolbar: Show Registry Activity, Show File System Activity.
  - The Show Network Activity and Show Process, and Threads Activity options can be disabled.



# Example1-Track File and Registry Changes

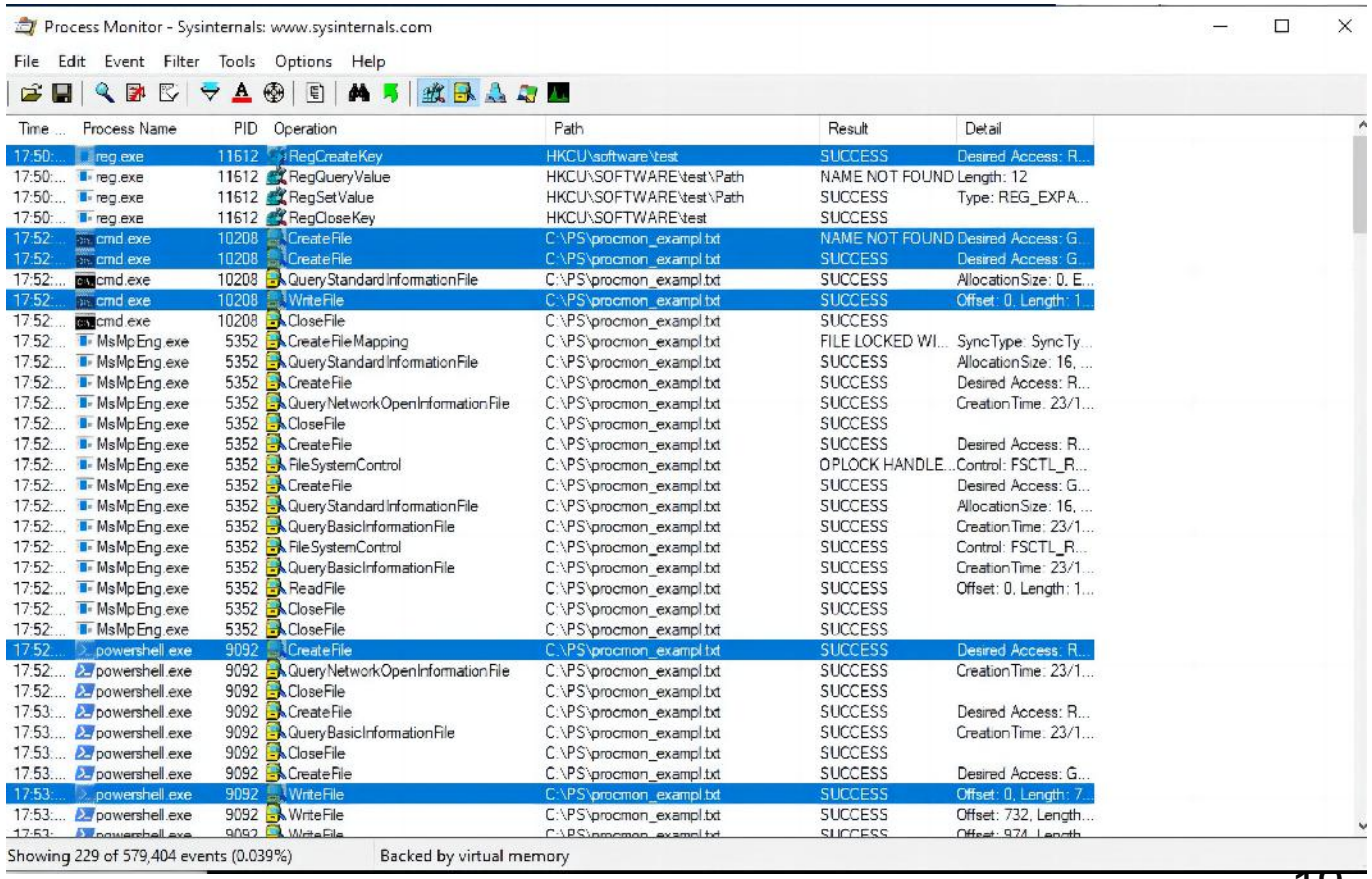
Let's say, you need to track access to the registry key HKEY_CURRENT_USER\Software\test and file c:\ps\procmon_example.txt.

- Step 8. Start event monitoring File > Capture Event.
- Step 9. Let's create a reg parameter key in the specified registry key using the command prompt:
  - reg add hkcu\software\test /v Path /t REG_EXPAND_SZ /d ^%systemroot^%
- Step 10. Let's write some data into the procmon_example.txt file using the command prompt:
  - echo %date%>>c:\ps\procmon_example.txt
- Step 11. And using PowerShell:
  - Get-Process|out-file C:\ps\procmon_example.txt

# Example1-Track File and Registry Changes

Let's say, you need to track access to the registry key HKEY_CURRENT_USER\Software\test and file c:\ps\procmon_example.txt.

- It contains events for creating a registry key by the reg.exe process (Operation > RegCreateKey).
- It also contains events of creation (Create File) and writing to a file (WriteFile) by the processes cmd.exe and powershell.exe.

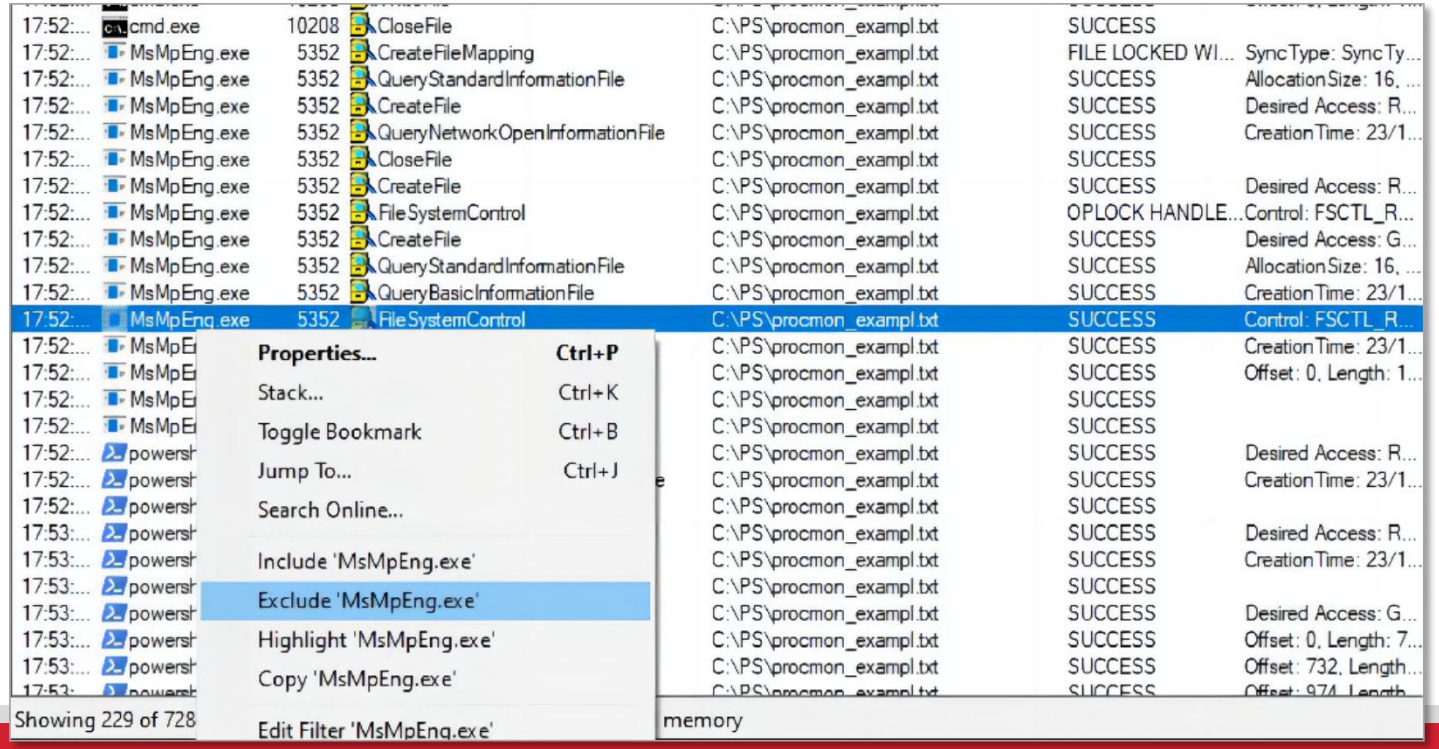



| Time ... | Process Name | PID | Operation | Path | Result | Detail |
|---|---|---|---|---|---|---|
| 17:50:... | reg.exe | 11612 | RegCreateKey | HKCU\software\test | SUCCESS | Desired Access: R... |
| 17:50:... | reg.exe | 11612 | RegQueryValue | HKCU\SOFTWARE\test\Path | NAME NOT FOUND | Length: 12 |
| 17:50:... | reg.exe | 11612 | RegSetValue | HKCU\SOFTWARE\test\Path | SUCCESS | Type: REG_EXPA... |
| 17:50:... | reg.exe | 11612 | RegCloseKey | HKCU\SOFTWARE\test | SUCCESS | |
| 17:52:... | cmd.exe | 10208 | CreateFile | C:\PS\procmon_exampl.txt | NAME NOT FOUND | Desired Access: G... |
| 17:52:... | cmd.exe | 10208 | CreateFile | C:\PS\procmon_exampl.txt | SUCCESS | Desired Access: G... |
| 17:52:... | cmd.exe | 10208 | QueryStandardInformationFile | C:\PS\procmon_exampl.txt | SUCCESS | AllocationSize: 0, E... |
| 17:52:... | cmd.exe | 10208 | WriteFile | C:\PS\procmon_exampl.txt | SUCCESS | Offset: 0, Length: 1... |
| 17:52:... | cmd.exe | 10208 | CloseFile | C:\PS\procmon_exampl.txt | SUCCESS | |
| 17:52:... | MsMpEng.exe | 5352 | CreateFileMapping | C:\PS\procmon_exampl.txt | FILE LOCKED WI... | SyncType: SyncTy... |
| 17:52:... | MsMpEng.exe | 5352 | QueryStandardInformationFile | C:\PS\procmon_exampl.txt | SUCCESS | AllocationSize: 16, ... |
| 17:52:... | MsMpEng.exe | 5352 | CreateFile | C:\PS\procmon_exampl.txt | SUCCESS | Desired Access: R... |
| 17:52:... | MsMpEng.exe | 5352 | QueryNetworkOpenInformationFile | C:\PS\procmon_exampl.txt | SUCCESS | CreationTime: 23/1... |
| 17:52:... | MsMpEng.exe | 5352 | CloseFile | C:\PS\procmon_exampl.txt | SUCCESS | |
| 17:52:... | MsMpEng.exe | 5352 | CreateFile | C:\PS\procmon_exampl.txt | SUCCESS | Desired Access: R... |
| 17:52:... | MsMpEng.exe | 5352 | FileSystemControl | C:\PS\procmon_exampl.txt | OPLOCK HANDLE... | Control: FSCTL_R... |
| 17:52:... | MsMpEng.exe | 5352 | CreateFile | C:\PS\procmon_exampl.txt | SUCCESS | Desired Access: G... |
| 17:52:... | MsMpEng.exe | 5352 | QueryStandardInformationFile | C:\PS\procmon_exampl.txt | SUCCESS | AllocationSize: 16, ... |
| 17:52:... | MsMpEng.exe | 5352 | QueryBasicInformationFile | C:\PS\procmon_exampl.txt | SUCCESS | CreationTime: 23/1... |
| 17:52:... | MsMpEng.exe | 5352 | FileSystemControl | C:\PS\procmon_exampl.txt | SUCCESS | Control: FSCTL_R... |
| 17:52:... | MsMpEng.exe | 5352 | QueryBasicInformationFile | C:\PS\procmon_exampl.txt | SUCCESS | CreationTime: 23/1... |
| 17:52:... | MsMpEng.exe | 5352 | ReadFile | C:\PS\procmon_exampl.txt | SUCCESS | Offset: 0, Length: 1... |
| 17:52:... | MsMpEng.exe | 5352 | CloseFile | C:\PS\procmon_exampl.txt | SUCCESS | |
| 17:52:... | MsMpEng.exe | 5352 | CloseFile | C:\PS\procmon_exampl.txt | SUCCESS | |
| 17:52:... | powershell.exe | 9092 | CreateFile | C:\PS\procmon_exampl.txt | SUCCESS | Desired Access: R... |
| 17:52:... | powershell.exe | 9092 | QueryNetworkOpenInformationFile | C:\PS\procmon_exampl.txt | SUCCESS | CreationTime: 23/1... |
| 17:52:... | powershell.exe | 9092 | CloseFile | C:\PS\procmon_exampl.txt | SUCCESS | |
| 17:53:... | powershell.exe | 9092 | CreateFile | C:\PS\procmon_exampl.txt | SUCCESS | Desired Access: R... |
| 17:53:... | powershell.exe | 9092 | QueryBasicInformationFile | C:\PS\procmon_exampl.txt | SUCCESS | CreationTime: 23/1... |
| 17:53:... | powershell.exe | 9092 | CloseFile | C:\PS\procmon_exampl.txt | SUCCESS | |
| 17:53:... | powershell.exe | 9092 | CreateFile | C:\PS\procmon_exampl.txt | SUCCESS | Desired Access: G... |
| 17:53:... | powershell.exe | 9092 | WriteFile | C:\PS\procmon_exampl.txt | SUCCESS | Offset: 0, Length: 7... |
| 17:53:... | powershell.exe | 9092 | WriteFile | C:\PS\procmon_exampl.txt | SUCCESS | Offset: 732, Length... |

# Example2-Exclude System Process

- Let's say, exclude msmpeng.exe (Antimalware Service Executable). This is the core process of the antimalware detection engine in Windows Defender.

- To exclude the events of this process from the ProcMon log, right-click on the process name msmpeng.exe and select Exclude.



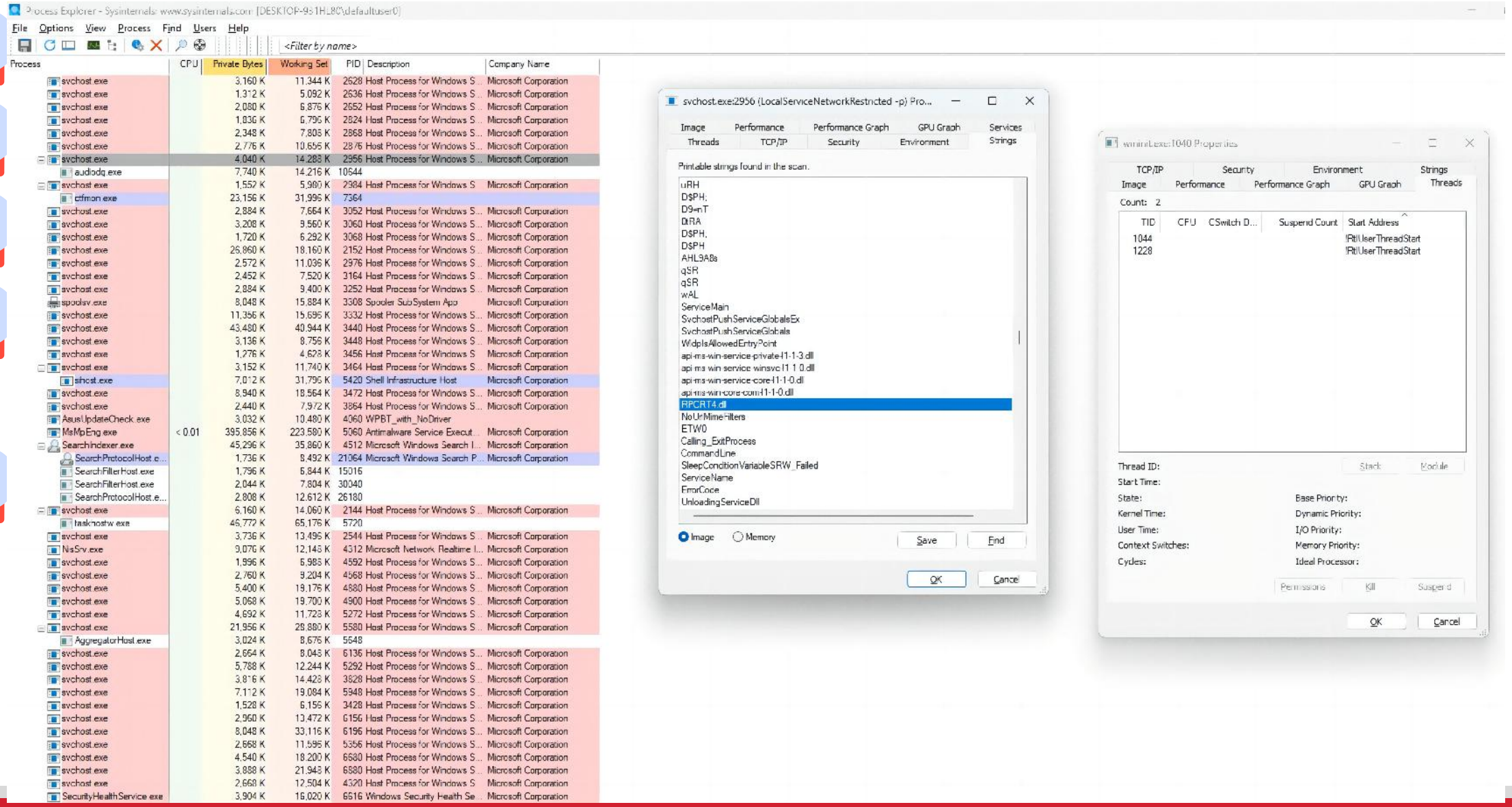
# Example2-Exclude System Process

- Let's say, exclude msmpeng.exe (Antimalware Service Executable). This is the core process of the antimalware detection engine in Windows Defender.

- This process will be added to the ProcMon filter with the Exclude value. It means that the ProcMon log won't display any activity from this process.

| | | | |
|---|---|---|---|
| Process Name | is | Procexp64.exe | Exclude |
| Process Name | is | System | Exclude |
| Process Name | is | MsMpEng.exe | Exclude |
| Operation | begins with | IRP_MJ_ | Exclude |
| Operation | begins with | FASTIO_ | Exclude |

# Process Explorer



# Process Explorer Vs. Process Moniter

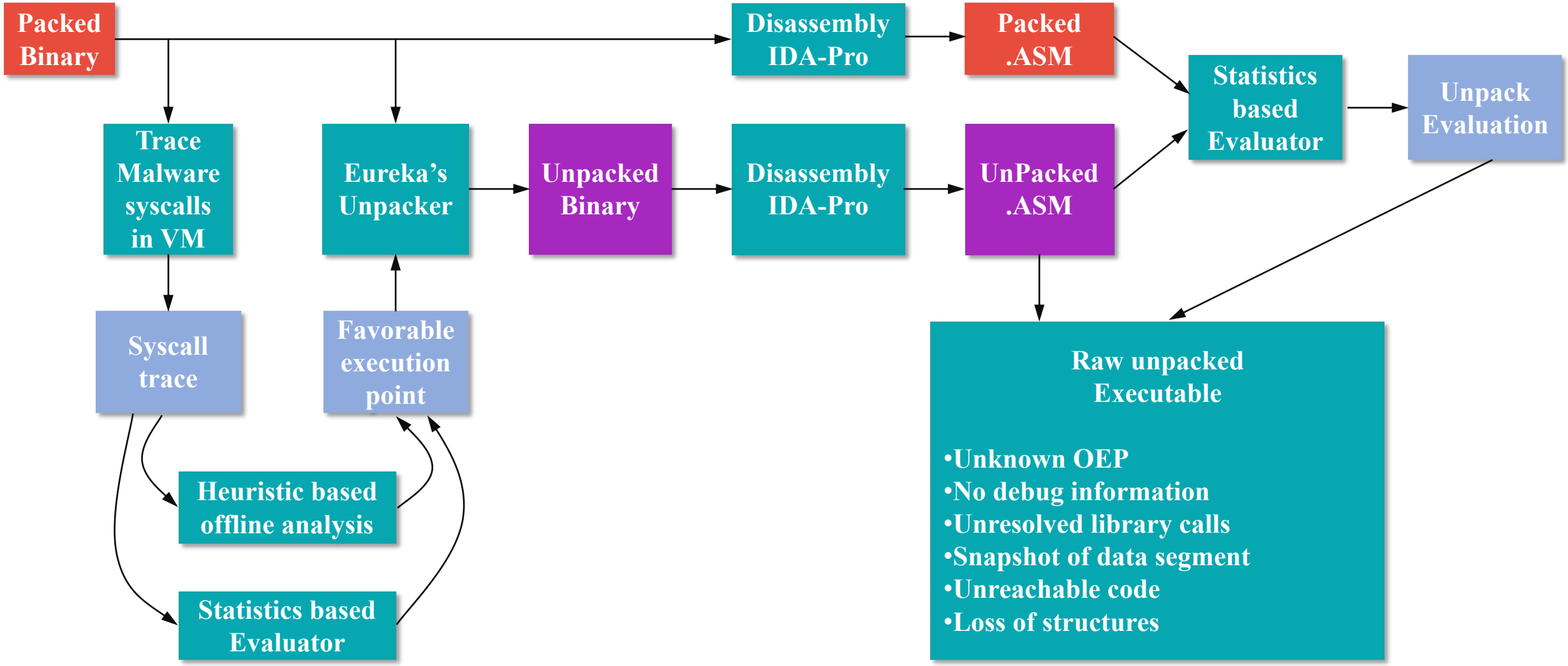
## Process Explorer

- Shows current state of each process
- Shows files, registry keys and thread loaded by each running process

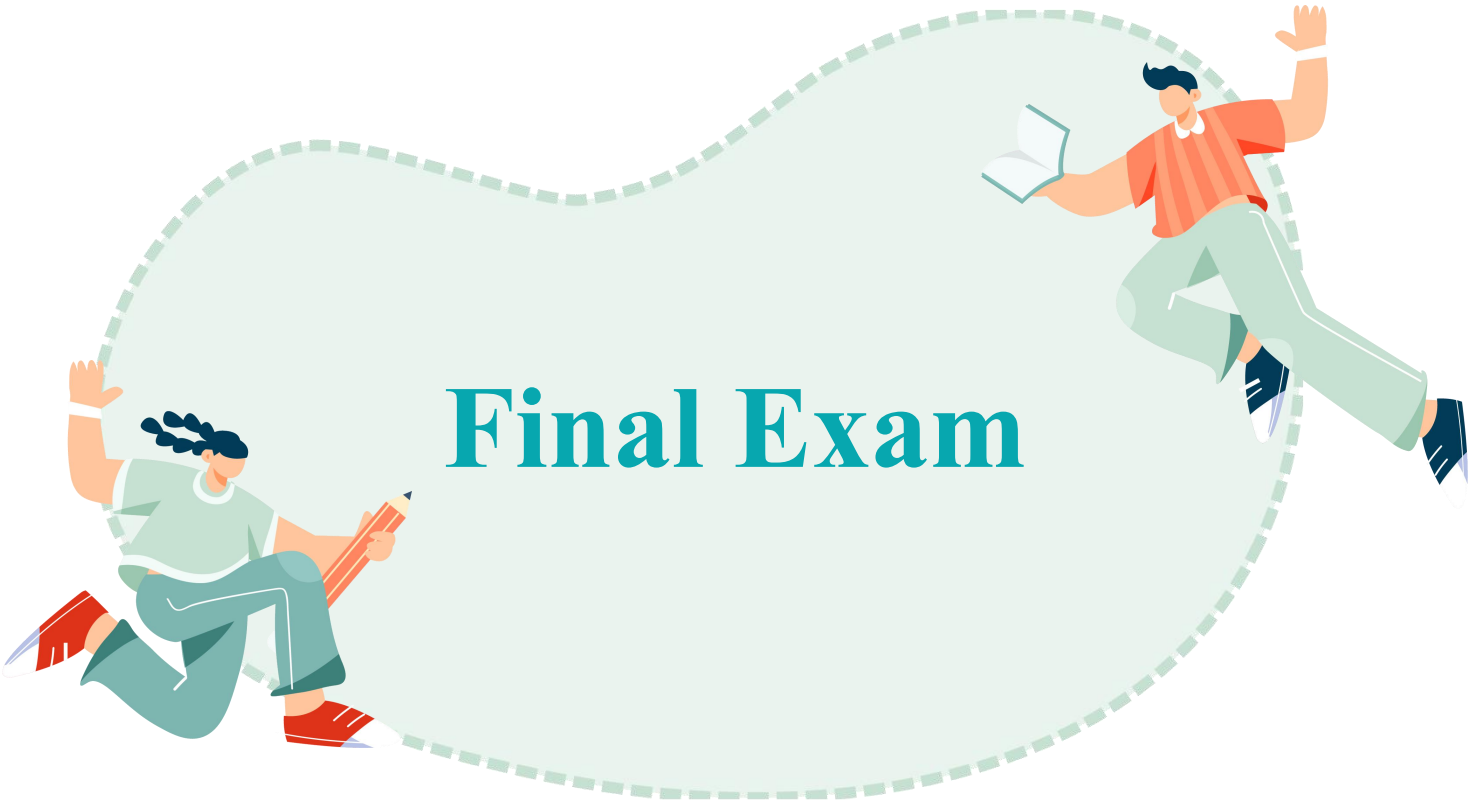
## Process Moniter

- In addition to monitoring, it logs process information- all events
- Logs show the file, registry, network, etc. the process attempted to use - successful or not
- “Access Denied” events also appear

# The Eureka Workflow



# Questions



# THE END

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